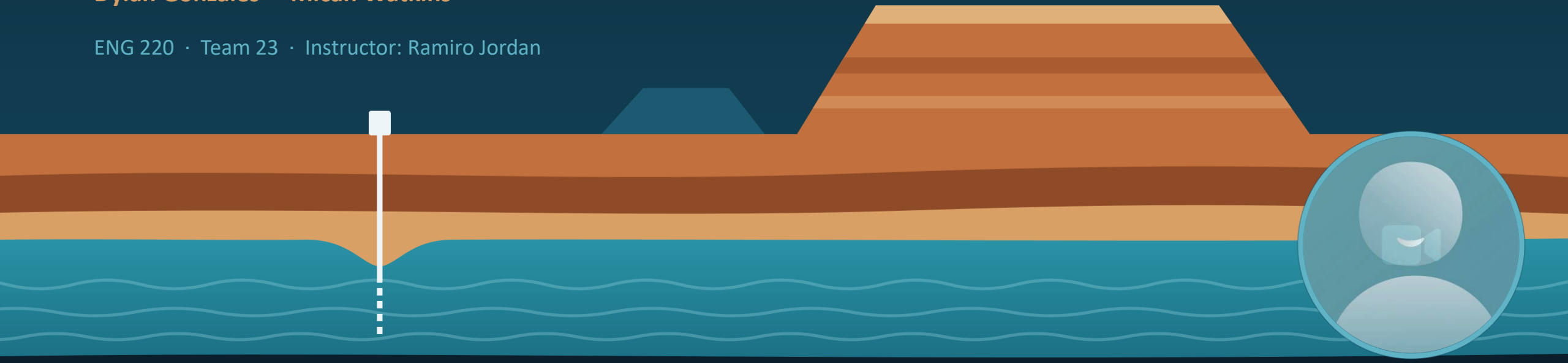
Beneath the Mesa

Assessing Aquifer and Well Water Resources Across Sandoval County, New Mexico

Petar Djordjevic (PM) · Meckrem Yousef · James Wauneka
Dylan Gonzales · Micah Watkins

ENG 220 · Team 23 · Instructor: Ramiro Jordan



The project at a glance

What we're building

- A screening study of water resources across Sandoval County, with Cuba as the focus
- An interactive app, a written report, and an open GitHub repository

What it does

- Brings together public well, aquifer, and river data
- Shows what's known about Cuba's groundwater, and where the data runs out

Who benefits

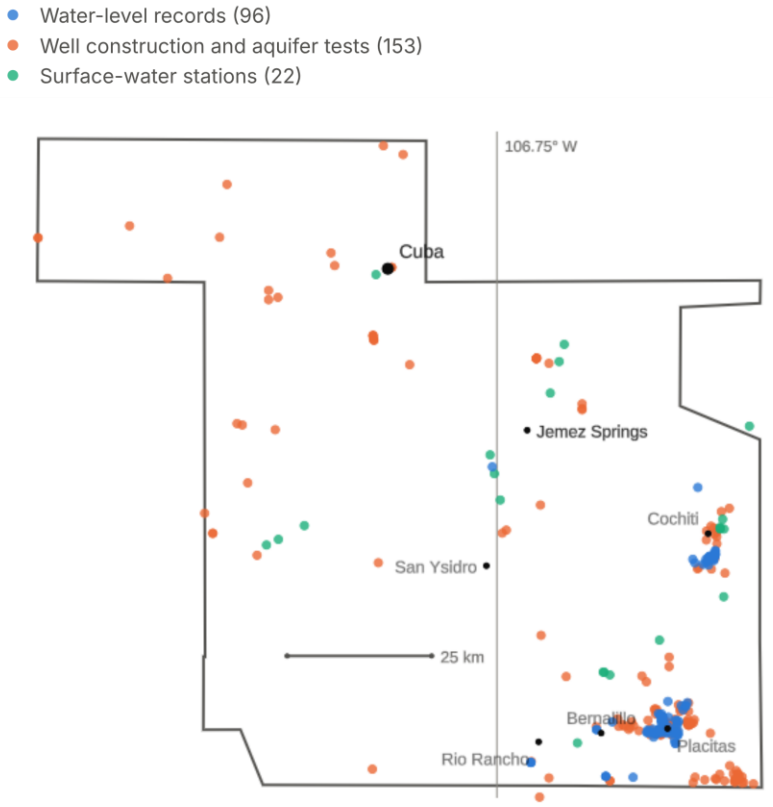
- Cuba residents and village leaders
- Rural communities that rely on wells
- Agencies deciding where to monitor next

Peace engineering
Clear, shared information about a scarce resource helps communities make fair water decisions together.



Background

- Where Sandoval County's water comes from: wells, rivers, and acequias
- Why the county spans several aquifer systems
- The concern: declining groundwater near Cuba
- What this project set out to check



Every site in the project's data (271), by type; Cuba is at upper left. Outline: US Census Bureau.

Research questions

MAIN QUESTION

Is Cuba's aquifer running out, and do public records say enough to tell?

1. **Groundwater levels**
   How have groundwater levels changed at monitored wells in Sandoval County?

2. **Cuba's aquifers**
   How do the Ojo Alamo Sandstone and San Jose Formation compare near Cuba?

3. **Coverage and data gaps**
   Where is monitoring coverage strong, and where are the biggest data gaps?



Deliverables

Deliverable	What it is	Owner	Due
Streamlit app	Interactive maps and charts of the data	Meckrem	**Nov 14**
Final report	Methods, findings, and figures in writing	Petar	**Nov 15**
Presentation slides	PowerPoint submitted on Canvas	Petar	**Nov 14**
Team video	Recorded presentation, each of us on camera	Everyone	**Nov 14**
Documentation site	How each stage was done, with links to everything	Petar	**Nov 15**
GitHub repository	Raw and cleaned data, scripts, and app code	Everyone	**Nov 15**

FINAL DUE

Mon, Nov 16

Sat, Nov 14
App, slides, and video done

Sun, Nov 15
Report and docs done

Mon, Nov 16, 12 PM
Video and slides on Canvas



Data

Groundwater levels

USGS: 9 wells, 1998–2026. NM Bureau of Geology: 87 wells, 1997–2019.

Aquifer tests

NM State Engineer: 179 tests from 1930 to 2008

Rio Puerco at Cuba

USGS gage 08332525: 10 streamflow visits from 2019 to 2026

Well logs and geology

NM State Engineer: 6 well logs. USGS and NMGS maps of the two formations.

Full source links are on the last slide and in the project documentation.



Methods

Collect

Dylan Gonzales

Seven public datasets, each with a one-page brief

Clean

James Wauneka

MATLAB script that puts the data in one shared format

Analyze

Micah Watkins

Trend statistics and 15 charts with labeled findings

Present

Meckrem Yousef

Streamlit app to explore the charts and data

Report

Petar Djordjevic

Report, slides, and documentation site

Tools Python · pandas · Plotly · MATLAB · Streamlit · GitHub



Data collection

Dylan Gonzales

Data acquisition and dataset justification

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

A data source page or the raw dataset

Figure caption and source

Data cleaning

James Wauneka

Data cleaning and preprocessing

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

The cleaning script, or the data before and after

Figure caption and source

Analysis and visualizations

Micah Watkins

Analysis code and visualizations

What I did

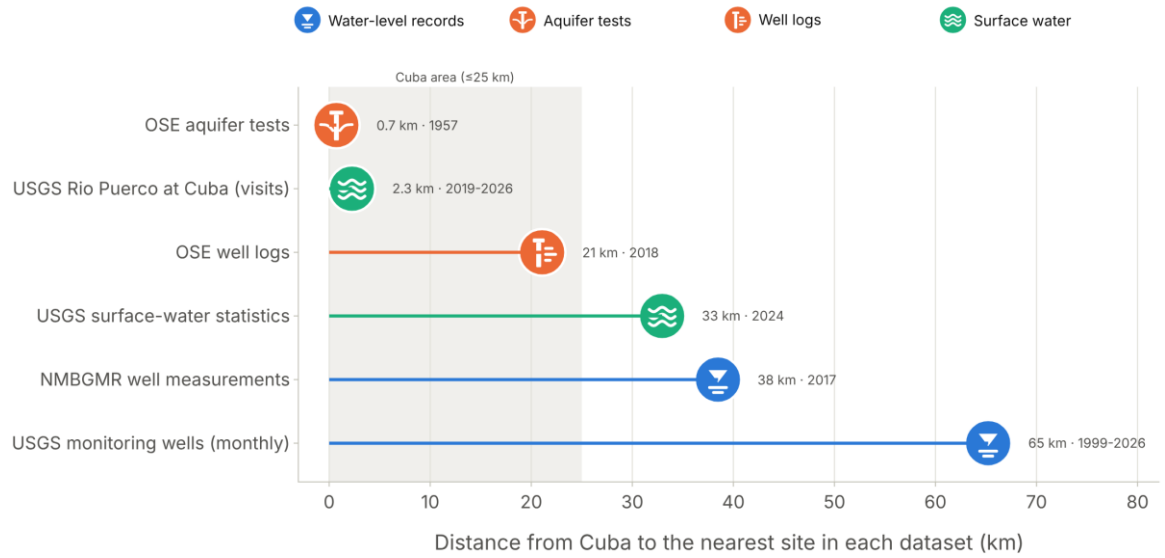
Turned the cleaned data into 15 charts on the six research questions, used in the app, report, and docs.

How I did it

Python, pandas, and Plotly; one function per chart returns the figure, caption, scope, and labeled findings.

What I learned

Thin data needs honest charts: almost nothing measures Cuba's groundwater, so each chart states its scope.



Nearest record of each kind to Cuba; only one-time tests lie within 25 km. Data: USGS, NMBGMR, NM OSE.

Streamlit app

Meckrem Yousef

Streamlit app design and development

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

A page of the live app

Figure caption and source

Report and documentation

Petar Djordjevic

Prime Minister: GitHub, report, slides, and video

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

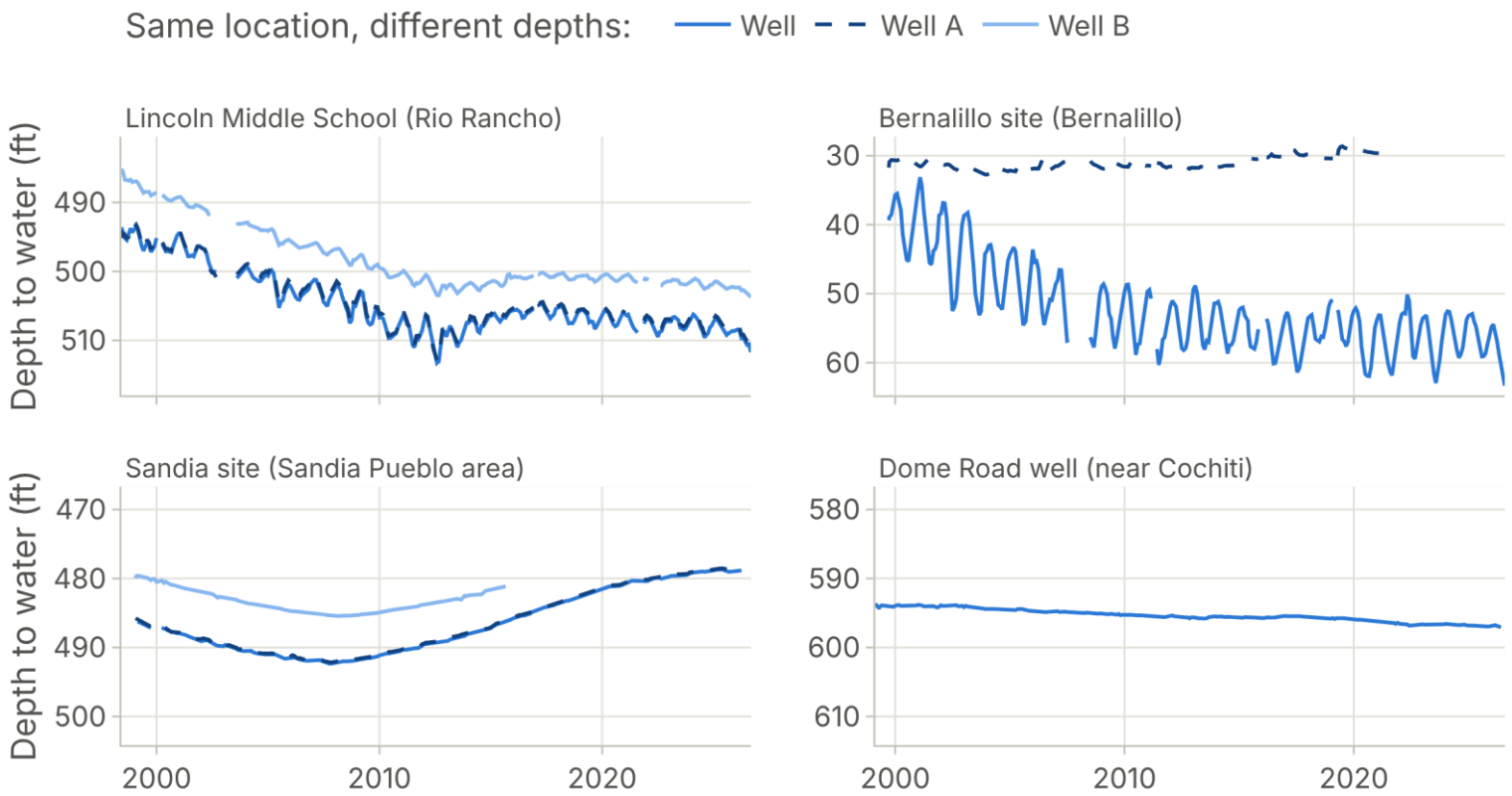
What I learned
A challenge you hit and how you handled it

Screenshot of your work

The documentation site or a page of the report

Figure caption and source

Most monitored wells fell



Monthly depth to water at the 9 USGS monitoring wells (4 sites), 1998–2026; axis flipped, so up means a higher water table. Source: USGS.

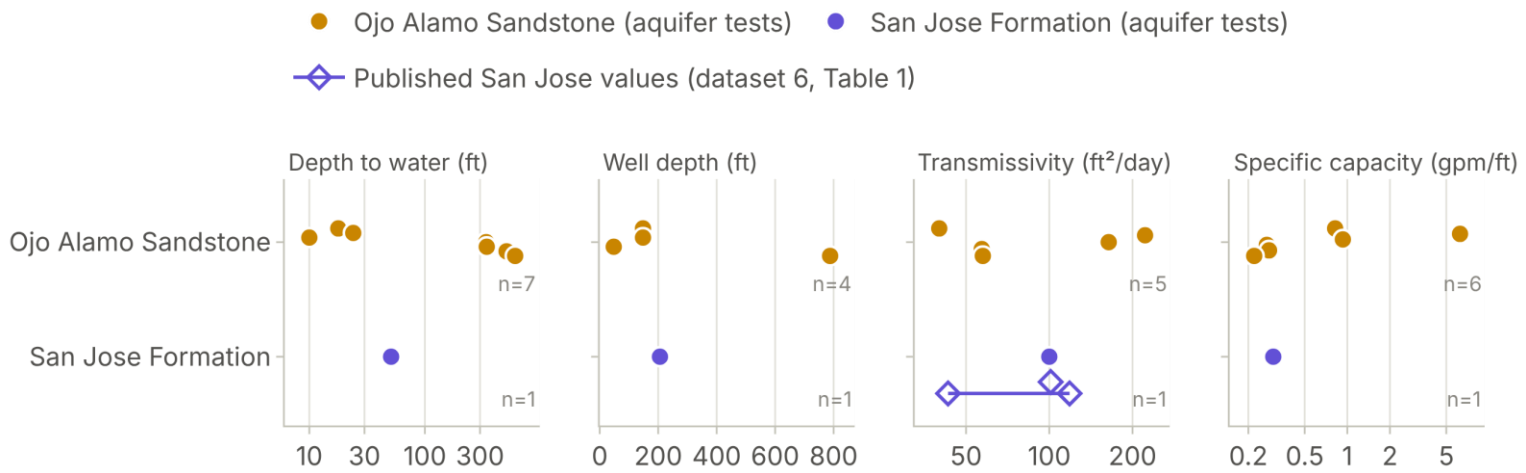
5 of 9

USGS monitoring wells where the water table fell, 1998–2026

What this shows

Three rose, including the Sandia site, up about 13 ft since 2007. All nine are on the county's east side, 65–95 km from Cuba.

Two aquifers, too few tests



Every aquifer test naming either formation; open diamonds are published San Jose values compiled by Blandford et al. (2025). Source: NM OSE.

7 vs 1

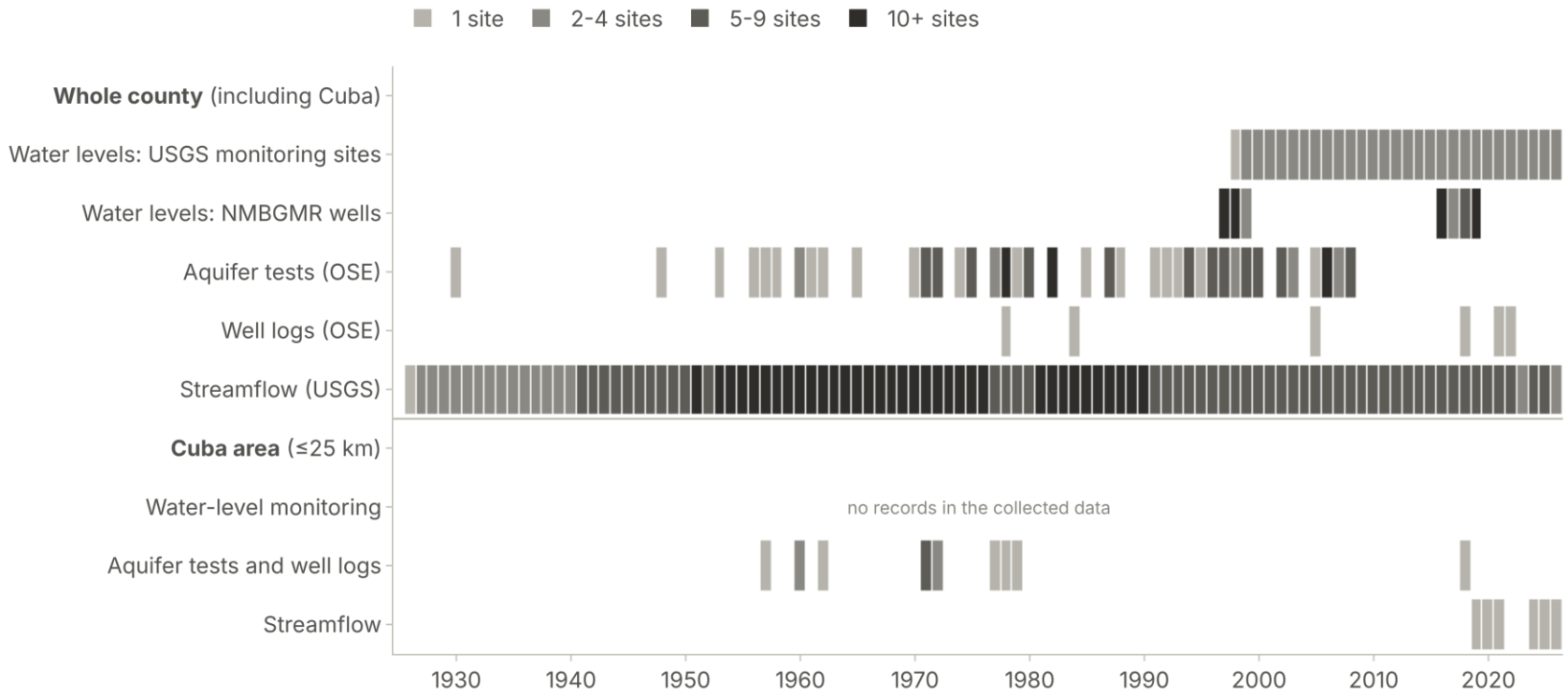
Aquifer-test events naming the Ojo Alamo vs. the San Jose, 1953–1977

What this shows

Too few to describe either formation, let alone compare them. Near Cuba, Ojo Alamo wells had water 18–24 ft down in 1957 and 1960.



Only 4 sites monitored yearly



Sites with at least one record each year, 1926–2026: whole county (top) and within 25 km of Cuba (bottom). Sources: USGS, NMBGMR, NM OSE.

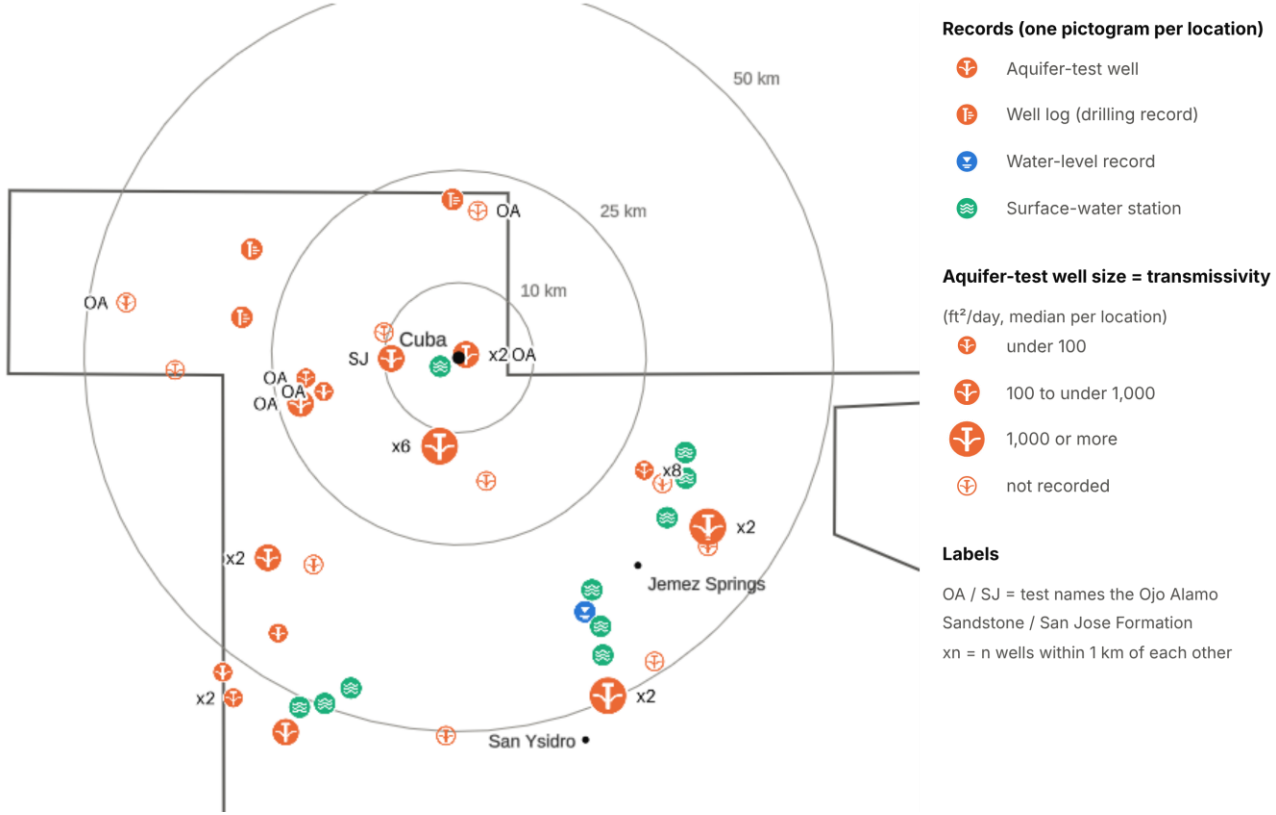
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Groundwater-level sites measured every year, all since 1999 and in the east

What this shows

Bureau of Geology wells: only 1997–1999 and 2016–2019. Near Cuba, the only recent records are river visits.

No monitoring wells near Cuba



Sites within 60 km of Cuba by type; rings at 10, 25, and 50 km. Sources: USGS, NMBGMR, NM OSE.

0

Water-level monitoring wells within 25 km of Cuba

What this shows

Near Cuba, 15 wells had one-time aquifer tests in 1957–1979. The nearest repeated water-level record is 38 km away.

Findings

What the data show

- 5 of 9 monitored wells fell, 1998–2026
- No monitoring well near Cuba

What they suggest

- Sandia's rise may reflect less pumping in Albuquerque
- Shallow Cuba wells can reach the Ojo Alamo

Too limited to tell

- Whether Cuba's aquifer is declining
- Which formation Cuba's wells draw from

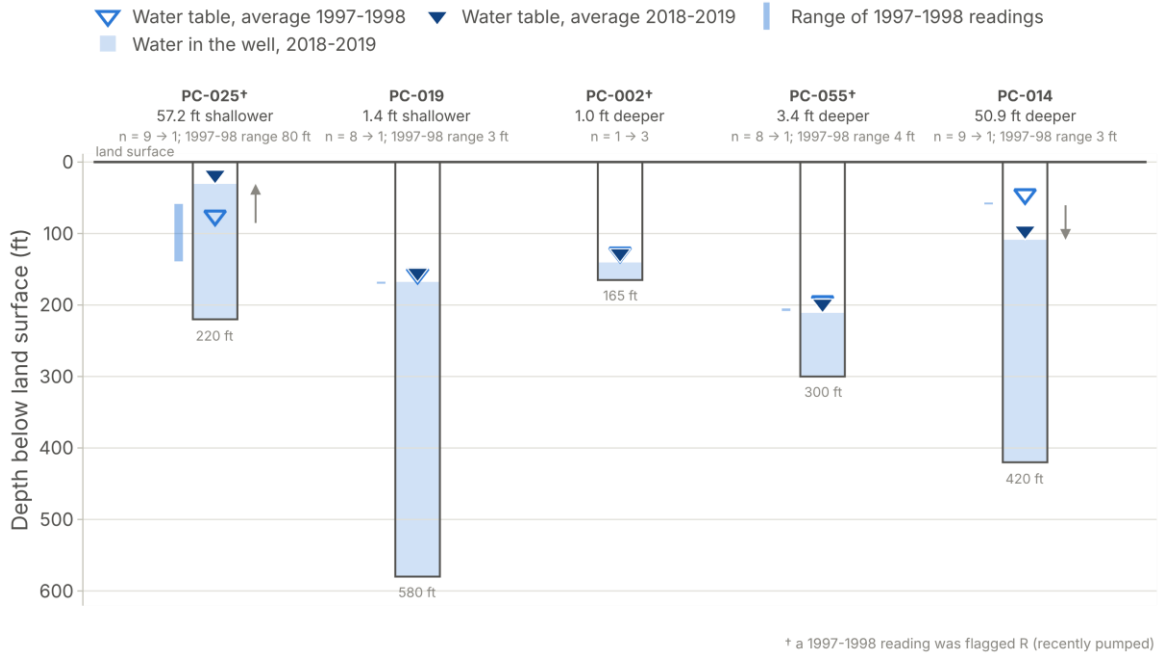
Bottom line

Public records can't tell whether Cuba's aquifer is running out: no well near Cuba has been monitored over time. Answering it needs the village's own well records.



Limitations

- No water-level monitoring within 25 km of Cuba; the 9 long records are 65–95 km away
- 82 of 87 Bureau of Geology wells were measured in only one period
- Not obtained: Cuba's municipal well records (formation, pumping, water levels)
- Needed: repeat water levels near Cuba and new aquifer tests in both formations



Only 5 of 87 Bureau of Geology wells were measured in both 1997–98 and 2018–19. Source: NMBGMR.

Risks and how we reduce them

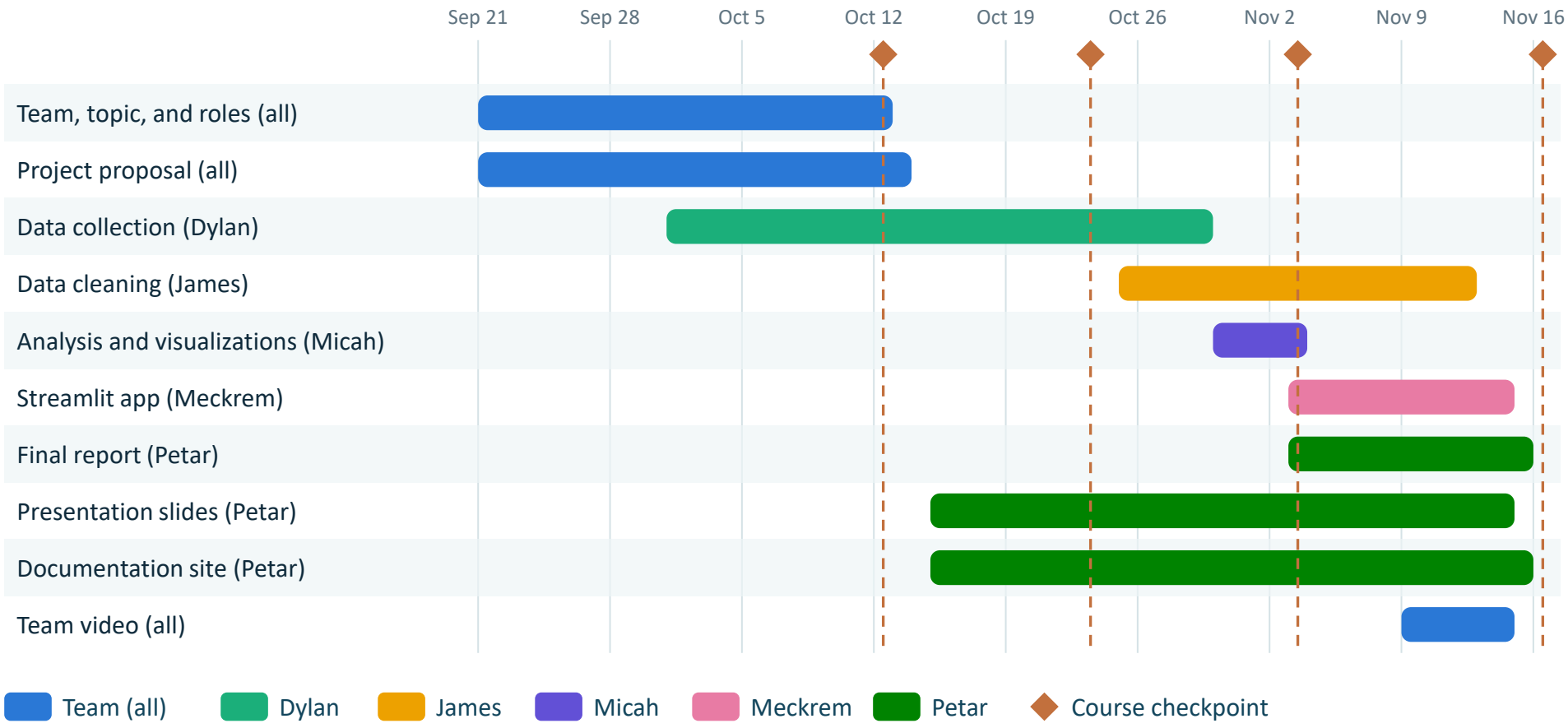
Risk	How we reduce it
No records from Cuba's own wells	Ask the Village of Cuba and NM OSE for records; label every result as Cuba, county, or regional
Sparse, uneven monitoring	Map where coverage is thin, trend only wells with long records, and state the gaps
Errors in the raw data	Shared cleaning rules, a rerunnable script, and checks against the raw files
Overstating conclusions	Keep what the data show, suggest, and can't tell separate in every finding
Tight final weeks	Write the report and slides alongside the app, and track every deadline on the timeline

Guardrails from our proposal

- Treat the concern about Cuba's aquifer as a question, not a conclusion
- Keep Cuba as the main case study to stay on schedule
- Use public data anyone can recheck



Timeline and milestones



Milestones

- **Mon, Oct 12**
  First quarter · done
- **Fri, Oct 23**
  Midpoint · done
- **Tue, Nov 3**
  Three-quarter · done
- **Mon, Nov 16**
  Final due



Explore the data, code, and app

GitHub repository
github.com/pdjordj11/eng_220_aquifer_project

Streamlit app
eng220aquiferproject.streamlit.app

Documentation
pdjordj11.github.io/eng_220_aquifer_project

Datasets
github.com/pdjordj11/eng_220_aquifer_project/tree/main/data

